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System Requirements Document

Abstract

N-MON is a compact, intelligent GNSS monitoring system designed to detect and classify jamming attacks targeting high-value cargo during transit. The project addresses the critical vulnerability of GNSS based fleet tracking systems, which are increasingly exploited by organized criminals using low cost jammers to steal goods worth millions.

N-MON provides a low cost, autonomous solution that combines real-time signal processing with Neural Network based classification. The system continuously monitors L1-band GNSS signals, extracts time-frequency representations, and employs a trained model to identify at least five interference types across different JSR levels. Upon detection, N-MON immediately alerts both the on-board user interface and a remote control center, enabling rapid response to protect assets.

Requirements list

USER REQUIREMENTS	
UREQ_1	The user shall be able to deploy and operate the system simply by connecting power and network, without complex configuration procedures
UREQ_2	The Control Center shall have a simple, intuitive way to see if GNSS signals are clean or compromised, and what type of interference is disrupting them
UREQ_3	Automatic warning shall be delivered to the user when jamming is detected

FUNCTIONAL REQUIREMENTS	
FREQ_1	The system shall continuously monitor GNSS signals (L band) multi-constellation
FREQ_2	The system shall acquire and then process GNSS signals to extract time-frequency representations
FREQ_3	The system shall integrate Artificial Intelligence to correctly identify and classify jamming interference across multiple interference types with different power of jamming signals JSR
FREQ_4	The system shall provide the user a simple UI to visualize the output of the AI algorithm and signal status
FREQ_5	The system shall automatically send an immediate warning if jamming is present to both the UI and a control center automatically send an immediate warning if jamming is present to both the UI and a control center

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TECHNICAL REQUIREMENTS	
TREQ_1	Multi-band GNSS patch antenna suitable for vehicle-mounted applications, providing reliable signal reception in variable conditions
TREQ_2	An SDR-based RF front-end capable of capturing GNSS signals, with an ADC that provides sufficient sampling rate and resolution for high-quality GNSS signal processing
TREQ_3	computational unit supporting real time NN inference and signal processing operations
TREQ_4	signal processing pipeline to elaborate input data using STFT
TREQ_5	test environment with capability to generate synthetic jamming signals with different power of jamming signals JSR
TREQ_6	communication protocols to reliably transmit alerts from the device to remote control station
TREQ_7	web based UI providing real-time status updates
TREQ_8	target cost lower than current laboratory setup
TREQ_9	reduced dimensions compared to existing laboratory setup